



Jomo Kenyatta University of Agriculture and Technology

School of Business and Entrepreneurship

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**BACHELOR OF SCIENCE IN BUSINESS INFORMATION
TECHNOLOGY**

**HBC 2304
RESEARCH METHODS**

COURSE TEACHING NOTES

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Introduction

The search for knowledge and information has been with us since time immemorial and will remain true in all decades to come. Through life, we face many problems which need proven and workable solutions; all made possible by research. Research enables us to move from the known to the unknown.

What is Research?

Research is an activity that people undertake in order to find out things in a systematic way, thereby increasing their knowledge. “*Systematic way*” – meaning research is based on logical relationships and not just beliefs. This is achieved by an explanation of the methods of collecting data, why results obtained are meaningful and limitations associated with them. “*To find out things*” – suggests there are a number of reasons for your study. This means that you need to describe, explain, understand, criticize and analyse. Therefore, research **MUST** have *a very clear purpose; data is collected and interpreted systematically in order to achieve that purpose.*

Research comprises “*creative work undertaken on a systematic basis in order to increase the stock of knowledge, including knowledge of humans, culture and society, and the use of this stock of knowledge to devise new applications.*” It is used to establish or confirm facts, reaffirm the results of previous work, solve new or existing problems, support theorems, or develop new theories. A research project may also be an expansion on past work in the field. Research projects can be used to develop further knowledge on a topic, or in the example of a school research project, they can be used to further a student's research prowess to prepare them for future jobs or reports.

To test the validity of instruments, procedures, or experiments, research may replicate elements of prior projects or the project as a whole. *The primary purposes of basic research (as opposed to applied research) are documentation, discovery, interpretation, or the research and development of methods and systems for the advancement of human knowledge.* Approaches to research depend on epistemologies, which vary considerably both within and between humanities and sciences. There are several forms of research: scientific, humanities, artistic, economic, social, business, marketing, practitioner research, life, technological, etc.

Purpose of Research

The primary purpose of research is to systematically discover new knowledge and advance existing understanding by applying scientific procedures to uncover hidden truths and find answers to questions. This leads to the development of new theories, concepts, and solutions that benefit society through improved decision-making and practical applications.

Research seeks to find answers to questions through the application of scientific procedures. The focus of research is to find out something that has not been discovered yet.

- a) To gain familiarity with a phenomenon and achieve new insights into it; referred to as *exploratory research*.
- b) To portray accurately the characteristics of a particular individual, situation or group; referred to as *descriptive research*.
- c) To show why relationships, patterns and links occur; referred to as *explanatory research*.

Forms of Research

Research is a systematic inquiry used to describe, explain, predict or control some observed phenomenon - the research topic. Research can be classified into four main forms based on the specific purpose:

Based on the Purpose of Research

- a) *Basic Research*; Aims to expand fundamental knowledge and understanding of theories and concepts.
- b) *Applied Research*; Focuses on solving practical, real-world problems and finding practical solutions.
- c) *Exploratory Research*; Used to identify and define a problem, or to explore a topic for which more information is needed.
- d) *Constructive Research*; Tests theories and proposes solutions to problems, often leading to the creation of new models or frameworks.
- e) *Empirical Research*; Relies on observable and verifiable evidence (empirical evidence) to test a hypothesis or prove a theory's feasibility.

Based on the Nature of the Data (Methodology)

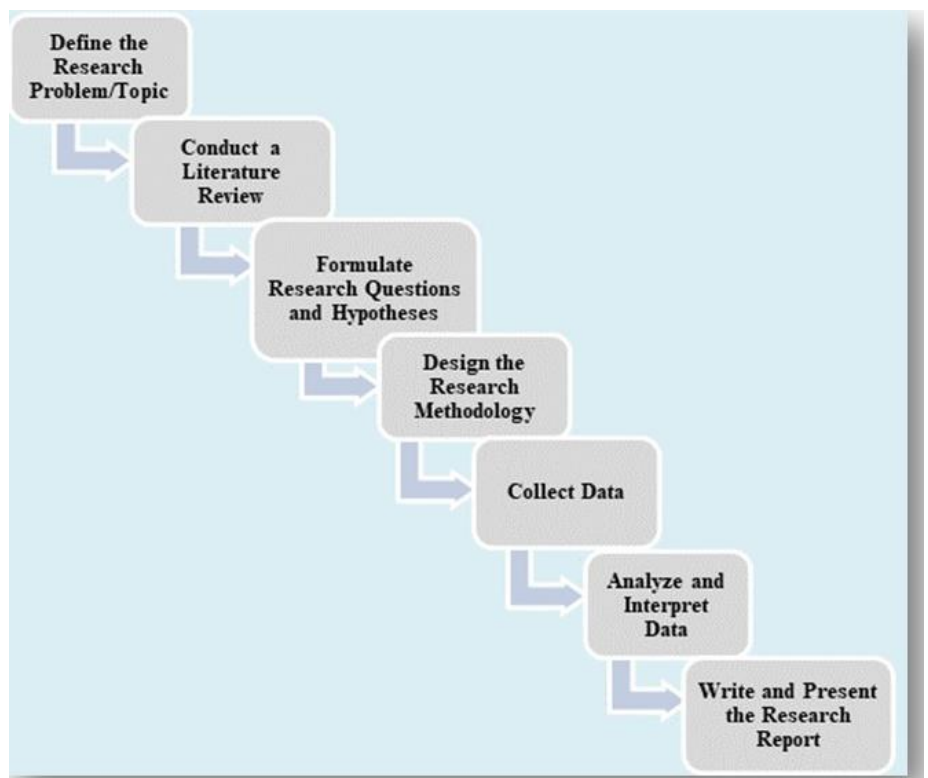
- i. *Qualitative Research*; Focuses on gaining an in-depth understanding of experiences, attitudes, and behaviors through non-numerical data, such as words, observations, or interviews.
- ii. *Quantitative Research*; Uses numerical data and statistical analysis to measure and quantify phenomena, often to test relationships between variables.
- iii. *Mixed Methods Research*; Combines both qualitative and quantitative approaches to provide a more comprehensive and nuanced understanding of a research question.

Based on the Research Design

- a. *Descriptive Research*; Aims to describe the characteristics of a population or phenomenon as it exists without intervening with the variables.
- b. *Correlational Research*; Examines the relationships between two or more variables, determining the extent to which they are associated.
- c. *Experimental Research*; Involves manipulating one or more independent variables to determine their effect on a dependent variable, establishing cause-and-effect relationships.
- d. *Causal-Comparative (Quasi-Experimental) Research*; Attempts to establish cause-and-effect relationships, but with less control over variables compared to true experimental research.

The Research Process

The research process is a systematic series of steps for conducting an investigation, beginning with identifying a problem or topic and narrowing it down, then conducting a literature review to understand existing knowledge. It moves through formulating specific research objectives, questions and hypothesis, designing the research methodology,



collecting, analyzing data and interpreting findings, and finally reporting the results. This structured approach allows for the generation of insightful, transparent, and impactful findings that advance knowledge and inform decision-making.

The process is often iterative, meaning you may revisit earlier steps as your understanding evolves. The key steps are:

1. *Define the Research Problem/Topic*; This involves identifying a specific issue or question that requires in-depth investigation to find a solution or gain new knowledge.
2. *Conduct a Literature Review*; Explore existing books, scholarly articles, and other relevant sources to understand what is already known about your topic, avoid duplication, and inform your research.
3. *Formulate Research Questions and Hypotheses*; Based on the literature review, develop specific questions to guide your research and create testable statements (hypotheses) about potential answers to those questions.
4. *Design the Research Methodology*; Plan how you will conduct the research, which includes selecting the appropriate study design, deciding on your sample (e.g., surveys or interviews), and choosing data collection methods.
5. *Collect Data*; Gather the necessary information or evidence from your chosen sample using the methods outlined in your research design.
6. *Analyze and Interpret Data*; Process the collected data using appropriate techniques and then interpret the findings to determine what they mean in relation to your research questions and hypotheses.
7. *Write and Present the Research Report*; Document your entire research process, including your methodology, findings, conclusions, and any recommendations, for others to review and use.

Primary Elements of a Research Project

1) Formulation of the Research Topic

Formulating and clarifying the research topic is the starting point in any research project. A viable research topic should focus to a certain set of ideas, persons or objects, a certain timeframe, a place often viewed through one or more related theories or viewpoints.

- *Example: “The Influence of Work-life Balance on Employee Productivity in Public Universities in Kenya”* might be a ‘researchable’ topic for study in human resource management. This is because it includes one specific HR practice and its influence on a particular HR consequence in a particular sector and country.

An associated theoretical literature may include various debates by subject matter experts, any published materials on the topic. You will benefit greatly from having conversations about your ideas with academics and colleagues who specialize in the area. Perhaps a good starting point is to focus on a subject or an area you know relatively well or what interests you most in your professional activity. It is always important to remember that the *best research comes out of deeply rooted curiosity, interest or passion and not some hope for short-term financial or career gain!* Think carefully about what really interests you and try to relate this with what you already know.

Features of a good research topic include:

- a) The topic should be something you are really fascinated about
- b) It should meet the examining body’s requirements and that it is at the correct level.
- c) The topic must be something you are capable of undertaking and one that excites your imagination.
- d) A good topic should easily allow definition of clear objectives and research questions.
- e) The topic should have ‘symmetry of outcomes’; that your results will be of similar value whatever you find out.
- f) It should be a topic that you already have research skills required to undertake it or you can develop the required skills within a short time
- g) The topic should be achievable within the available timeframe
- h) The project should be current by the time you finish it.
- i) The research topic should be achievable within the financial resources available for the purpose.

2) Identification and Definition of the Research Problem

A research problem is a definite or clear expression (statement) about an area of concern, a condition to be improved upon, a difficulty to be eliminated, or a troubling question that exists in scholarly literature, in theory, or within existing practice that point to a need for meaningful understanding and deliberate

investigation. A research problem does not state how to do something, offer a vague or broad proposition, or present a value question.

The identification of research problem is the first and foremost step that every researcher has to undertake. At times, it becomes rather difficult for an inexperienced researcher or a novice/beginner in research to conceptualize a research problem. In general, a research problem should be understood as some difficulty, unclear situation which a researcher experiences in practical or theoretical context and wants to obtain a tangible explanation, clarification or offer solution to it. For students, this problem may be as a result of theoretical encounter in the area of specialization. As such, before embarking on any research, you should identify the major research area of your interest, mostly the area of your specialization - Education, Social sciences, Humanities, Business administration among others.

Once you have the broad area, you narrow down the area by selecting a particular topic. This should be done after going through most of the literature related to the area. The topic should further be narrowed down to a specific researchable problem. In general, a research problem should be understood as some difficulty, unclear situation which a researcher experiences in practical or theoretical context and wants to obtain a tangible explanation, clarification or offer solution to it. For students, this problem may be as a result of theoretical encounter in the area of specialization. Before embarking on any research, you should identify the major research area of your interest, mostly the area of your specialization; Human Resource Management, IT, Leadership, Public Administration, Entrepreneurship, Strategic Management, Project Management, among others.

Components of a Research Problem

For a research problem to exist, there are a number of core elements that have to be inherent. There must be:

1. *An individual or community or an organization/institution to whom the problem could be attributed.* These occupy a certain geographical area. For instance, teacher/parental factors affecting students' performance in private secondary schools in Arusha Municipality. In this study, there are individuals (parents, teachers, students), there are institutions (private secondary schools), and there is the area of study (Arusha City).
2. *Some Objectives for pursuing the problem.* There must be some objectives pursuing the problem, otherwise it would be repugnant to reason and common understanding to undertake the research. For example: To find out teacher/parental factors affecting the students' academic performance in private secondary schools in Arusha City.
3. *Some lines of action to be taken.* There must be at least two lines of action to be taken to attain the objective. For example, poor academic performance may be attributed to negative teacher and parental factors. Thus altering negative teacher factors and parental factors become the lines of action to be pursued. Here, the underlying question is "*what is the cause of this problem – poor academic performance?*" It is in answering this question that you must pursue some lines of action through stating some variables (teacher factors and parental factors).

Criteria for Formulating the Problem

The selection of one appropriate researchable problem out of the identified problems requires evaluation of those alternatives against certain criteria, which may be grouped into:

1. Internal Criteria

- i. *Researcher's interest:* The problem should interest the researcher and be a challenge to him. Without interest and curiosity, he may not develop sustained perseverance. Interest in a problem depends upon the researcher's educational background, experience, outlook and sensitivity.
- ii. *Researcher's own resource:* In the case of research to be done by a researcher on his own, consideration of his own financial resource is pertinent. If it is beyond his means, he will not be able to complete the work, unless he gets some external financial support. Time resource is more important than finance. Research is a time-consuming process; hence it should be properly utilized.

- iii. *Researcher's competence:* A mere interest in a problem will not do. The researcher must be competent to plan and carry out a study of the problem. He must possess adequate knowledge of the subject-matter, relevant methodology and statistical procedures.

2. **External Criteria**

- i. *Research-ability of the problem:* The problem should be researchable, i.e., amendable for finding answers to the questions involved in it through the scientific method.
- ii. *Novelty of the problem:* The problem must have novelty. There is no use of wasting one's time and energy on a problem already studied thoroughly by others.
- iii. *Importance and urgency:* Problems requiring investigation are unlimited, but available research efforts are very much limited.
- iv. *Facilities:* Research requires certain facilities such, as well-equipped library facility, suitable and competent guidance, data analysis facility, etc. Hence the availability of the facilities relevant to the problem must be considered. Problems for research, their relative importance and significance should be considered.
- v. *Feasibility:* A problem may be a new one and also important, but if research on it is not feasible, it cannot be selected.
- vi. *Usefulness and social relevance:* Above all, the study of the problem should make a significant contribution to the concerned body of knowledge or to the solution of some significant practical problem. It should be socially relevant.
- vii. *Research personnel:* Research undertaken by professors and by research organizations require the services of investigators and research officers. But in India and other developing countries, research has not yet become a prospective profession. Hence talent persons are not attracted to research projects.

3) **Formulation of Research Objectives**

Research objectives represent the targets you seek to accomplish through the study. The relation of this section with the problem statement must be made clear and the section forms a basis for judging the rest of the proposal. Objectives are usually structured as general and specific, all emanating from the research topic and problem. The general objective is the aim/purpose of your research while specific objectives are specific outputs or results you accomplish to add up to your research goal/purpose.

Characteristics of the research objectives

1. *Clear;* Clarity is the main focus of the research objectives. The way you word your research objectives tell a lot about your research. The wording should have to be very clear and very well defined. Not only other people will get to know about your research from the research objectives but you too need to look at the objectives to make several decisions about your research. Make simple sentences but clear and avoid complexities in wording.
2. *Complete;* Each phrase in the research objectives should have a sense of completeness. Broken sentences and incomplete objectives can lead you into trouble later on. Here completeness is necessary because you will be able to decide the extent of your research from your research objectives. A complete research objective should have information about the population to be studied.
3. *Specific;* Your research objectives should specifically focus on the what, how when and where questions and there should be no loopholes.
4. *Identify the main variables of the study;* In the research objectives the researcher decides about the main variables of the study. The variables are the measurable concepts or quantities that will majorly impact your research. The variables should be well-defined in this phase of the research. You should decide about the dependent variables, independent variables as well as the control variables; if the research is experimental research.
5. *Identify the direction of the relationship;* Some research studies have hypothesis. In hypothesis testing studies the researcher should define the direction in which he wants to test the variables. This is only done in hypothesis testing studies; in other studies, the researcher only mentions the variables.

Research objectives MUST be operationalized, where both the independent and dependent variables must be expressly evident.

Example: To determine the influence of globalization (IV) on the performance (DV) of the medium-size juice processing firms in Kenya.

The objectives provide an accurate description of the specific actions you will take in order to reach the research purpose. As with the problem formulation, the overall objective should be framed in a single sentence.

4) Formulation of Research Questions/Hypotheses

Research hypotheses/questions as they relate to this chapter are broad propositions/questions which the study seeks to provide answers. As to the research hypotheses, they are stated in “null” form and the study seeks to accept or reject each one of them depending on the results. Research questions or hypotheses are directly formulated from the research objectives and MUST have same focus in content and intent – they must also be operationalized as is the case with research objectives. As a rule, if there are three objectives, research hypotheses or questions must also be three!

Example:

- ✓ *Question:* “How does beneficiary participation influence the performance of community development projects in Kisii County?”
- ✓ *Hypothesis:* “Beneficiary participation has no significant influence on the performance of community development projects in Kisii County.”

Depending on the study findings, a hypothesis is either rejected or accepted.

Literature Review

A literature review is an overview of the previously published works on a topic. It is a text of a scholarly paper, which includes the current knowledge including substantive findings, as well as theoretical and methodological contributions to a particular topic. Literature reviews are secondary sources, and do not report new or original experimental work. Most often associated with academic-oriented literature, such reviews are found in academic journals, and are not to be confused with book reviews that may also appear in the same publication.

Literature reviews are a basis for research in nearly every academic field. A narrow-scope literature review may be included as part of a peer-reviewed journal article presenting new research, serving to situate the current study within the body of the relevant literature and to provide context for the reader. In such a case, the review usually precedes the methodology and results sections of the work.

Purpose of the Literature Review

- a) It gives readers easy access to research on a particular topic by selecting high quality articles or studies that are relevant, meaningful, important and valid and summarizing them into one complete report
- b) It provides an excellent starting point for researchers beginning to do research in a new area by forcing them to summarize, evaluate, and compare original research in that specific area
- c) It ensures that researchers do not duplicate work that has already been done
- d) It can provide clues as to where future research is heading or recommend areas on which to focus
- e) It highlights key findings
- f) It identifies inconsistencies, gaps and contradictions in the literature
- g) It provides a constructive analysis of the methodologies and approaches of other researchers

Importance of Literature Review in Research

Doing a careful and thorough literature review is essential when you write about research at any level. It is basic homework that is assumed to have been done vigilantly, and a given fact in all research papers. By providing one, usually offered in your introduction before you reach your thesis statement, you are telling your reader that you have not neglected the basics of research.

It not only surveys what research has been done in the past on your topic, but it also appraises, encapsulates, compares and contrasts, and correlates various scholarly books, research articles, and other relevant sources that are directly related to your current research. Given the fundamental nature of providing one, your research paper will be not considered seriously if it is lacking one at the beginning of your paper.

Adequate and comprehensive literature review:

1. *Demonstrates that you know the field* - This means more than reporting what you've read and understood. Instead, you need to read it critically and to write in such a way that shows you have a feel for the area; you know what the most important issues are and their relevance to your work, you know the controversies, you know what's neglected, you have the anticipation of where it's being taken. All this would allow you to map the field and position your research within the context.
2. *Justifies the reason for your research* - This is closely connected with demonstrating that you know the field. It is the knowledge of your field which allows you to identify the gap which your research could fill. However, it is not enough to find a gap. You have also to be able to convince your reader that what you are doing is important and needs to be done.
3. *Allows you to establish your theoretical framework and methodological focus*. Even if you are proposing a new theory or a new method, you are doing so in relation to what has been done.
4. *Creates a Rapport with Your Audience* - A literature review helps you create a sense of rapport with your audience or readers so they can trust that you have done your homework. As a result, they can give you credit for your due diligence: you have done your fact-finding and fact-checking mission, one of the initial steps of any research writing. As a student, you may not be an expert in a given field; however, by listing a thorough review in your research paper, you are telling the audience, in essence, that you know what you are talking about. As a result, the more books, articles, and other sources you can list in the literature review, the more trustworthy your scholarship and expertise will be.
5. *Helps You Avoid Incidental Plagiarism* - Imagine this scenario; You have written a research paper, an original paper in your area of specialization, without a literature review. When you are about to publish the paper, you soon learn that someone has already published a paper on a topic very similar to yours. Of course, you have not plagiarized anything from that publication; however, if and when you publish your work, people will be suspicious of your authenticity. They will ask further about the significance of repeating similar research. In short, you could have utilized the time, money, and other resources you have wasted on your research on something else. Had you prepared a literature review at the onset of your research, you could have easily avoided such mishap. During the compilation of your review, you could have noticed how someone else has done similar research on your topic. By knowing this fact, you can tailor or tweak your own research in such a way that it is not a mere rehashing of someone else's original or old idea.
6. *Sharpens Your Research Focus* - As you assemble outside sources, you will condense, evaluate, synthesize, and paraphrase the gist of outside sources in your own words. Through this process of winnowing, you will be able to place the relevance of your research in the larger context of what other researchers have already done on your topic in the past. The literature review will help you compare and contrast what you are doing in the historical context of the research as well as how your research is different or original from what others have done, helping you rationalize why you need to do this particular research. It is essential in helping you shape and guide your research in the direction you may not have thought of by offering insights and different perspectives on the research topic.

Types of Literature Review

Literature reviews are pervasive throughout various academic disciplines, and thus you can adopt various approaches to effectively organize and write your literature review. There are two main approaches to literature review.

a) Theoretical Review

The theoretical literature review help establish what theories already exist, the relationships between them, to what degree the existing theories have been investigated, and to develop new hypotheses to be tested. The purpose of this form is to concretely examine the corpus of theory that has accumulated in regard to

an issue, concept, theory, phenomena. The theoretical literature review help establish what theories already exist, the relationships between them, to what degree the existing theories have been investigated, and to develop new hypotheses to be tested. Often this form is used to help establish a lack of appropriate theories or reveal that current theories are inadequate for explaining new or emerging research problems. The unit of analysis can focus on a theoretical concept or a whole theory or framework.

In theoretical review, clearly describe the framework, concepts, models, or specific theories that underpin your study. This includes noting who the key theorists are in the field who have conducted research on the problem you are investigating and, when necessary, the historical context that supports the formulation of that theory.

b) Empirical Review

An empirical literature review process involves the evaluation of previous empirical studies to bring to rest a specific research issue. When conducting an empirical review, the researcher searches for empirical studies in a similar field, to uncover gaps, convergence, and conflicts that can help to properly identify, describe, and repeat the process. In the empirical review section of the literature review, the author examines the current state of knowledge and theories on the subject, as well as the topic's historical context.

In an ideal situation, an empirical review should provide information on the name of the author(s), the title of the paper, the number of objectives raised, and research questions asked, the study design, the population and sampling techniques, the data collection instrument as well as the data analysis procedures and techniques. The most significant findings of the research should also be highlighted as well as recommendations made.

At times providing all this information in a highly coherent manner is enough, however, some schools or supervisors may require you to critique the reviewed empirical studies further, especially by highlighting the strength and weaknesses of the various methodological apparatus, tools, and techniques adopted by the author (s) of the reviewed empirical studies.

Research Approaches

Research approaches are overarching strategies for conducting a study, including the collection, analysis, and interpretation of data, and are broadly categorized into quantitative, qualitative, and mixed-methods approaches. Quantitative research focuses on numerical data to find patterns and make generalizations, while qualitative research explores non-numerical data for deep insights into experiences, behaviors, and meanings. Mixed-methods research integrates both quantitative and qualitative approaches to provide a more comprehensive understanding of a phenomenon

There are three main approaches to research:

1. Qualitative Research

Qualitative Research is primarily exploratory research. It is used to gain an understanding of underlying reasons, opinions, and motivations. It provides insights into the problem or helps to develop ideas or hypotheses for potential quantitative research. Qualitative Research is also used to uncover trends in thought and opinions, and dive deeper into the problem. Qualitative data collection methods vary using unstructured or semi-structured techniques. Some common methods include focus groups (group discussions), individual interviews, and participation/observations. The sample size is typically small, and respondents are selected to fulfill a given quota.

Qualitative research gives the researcher more flexibility, allows one to take advantage of the richness of data and thus to obtain more meaningful results. It affords the opportunity to examine the process of "why" and "how", not just "what"; to explore the complex, interdependent issues that constitute international management. The "what" are the factor that need to considered; the "how" refers to how they are related; and the "why" are the underlying dynamics that justify the selection of those factors and their relationship.

Qualitative researchers do not base their research on pre-determined hypotheses. The approach to data collection and analysis is methodical but allows for greater flexibility than in quantitative research. Data is

collected in textual form on the basis of observation and interaction with the participants e.g. through participant observation, in-depth interviews and focus groups. It is not converted into numerical form and is not statistically analyzed.

2. Quantitative Research

Quantitative Research is used to quantify the problem by way of generating numerical data or data that can be transformed into useable statistics. It is used to quantify attitudes, opinions, behaviors, and other defined variables – and generalize results from a larger sample population. Quantitative Research uses measurable data to formulate facts and uncover patterns in research. Quantitative data collection methods are much more structured than Qualitative data collection methods. Quantitative data collection methods include various forms of surveys – online surveys, paper surveys, mobile surveys and kiosk surveys, face-to-face interviews, telephone interviews, longitudinal studies, website interceptors, online polls, and systematic observations.

Quantitative methodology, is applicable where the aim of the research is to ascertain how many, what and where. In seeking such answers, a quantitative approach relies on the use of predetermined response categories by means of standardized data collection instruments such as mail survey, or structured or semi-structured interviews so as to enable statistical techniques to be used to assist in the interpretation of data. The standardized measurement and sampling procedures are intended to enhance the reliability of observation, facilitate replication studies, and allow generalization to a larger population.

3. Pragmatic Approach to Research (mixed methods)

The pragmatic approach to science involves *using the method which appears best suited to the research problem and not getting caught up in philosophical debates about which is the best approach*. Pragmatic researchers therefore grant themselves the freedom to use any of the methods, techniques and procedures typically associated with quantitative or qualitative research. *They recognize that every method has its limitations and that the different approaches can be complementary.*

They may also use different techniques at the same time or one after the other. For example, they might start with face-to-face interviews with several people or have a focus group and then use the findings to construct a questionnaire to measure attitudes in a large-scale sample with the aim of carrying out statistical analysis. Depending on which measures have been used, the data collected is analyzed in the appropriate manner. However, it is sometimes possible to transform qualitative data into quantitative data and vice versa although transforming quantitative data into qualitative data is not very common.

Research Designs

A research design is the overall strategy and framework for a study, guiding decisions on research questions, data collection and analysis methods, and ensuring valid, reliable, and accurate results. It acts as a blueprint for research, defining the methods and techniques to systematically answer research questions, collect and interpret data, and provide structure and direction to the entire project.

The function of a research design is to ensure that the evidence obtained enables you to effectively address the research problem logically and as unambiguously as possible. In social sciences research, obtaining information relevant to the research problem generally entails specifying the type of evidence needed to test a theory, to evaluate a program, or to accurately describe and assess meaning related to an observable phenomenon. With this in mind, a common mistake made by researchers is that they begin their investigations far too early, before they have thought critically about what information is required to address the research problem. Without attending to these design issues beforehand, the overall research problem will not be adequately addressed and any conclusions drawn will run the risk of being weak and unconvincing. As a consequence, the overall validity of the study will be undermined.

1. Descriptive Research

What is going on (descriptive research)?

Descriptive research is used to describe characteristics of a population or phenomenon being studied. It does not answer questions about how/when/why the characteristics occurred. Rather it addresses the "what"

question (what are the characteristics of Minnesota state population or situation being studied?). The characteristics used to describe the situation or population is usually some kind of categorical scheme also known as descriptive categories. For example, the periodic table categorizes the elements. Scientists use knowledge about the nature of electrons, protons and neutrons to devise this categorical scheme. We now take for granted the periodic table, yet it took descriptive research to devise it. Descriptive research generally precedes explanatory research. For example, over time the periodic table's description of the elements allowed scientists to explain chemical reaction and make sound prediction when elements were combined.

Hence, descriptive research cannot describe what caused a situation. Thus, descriptive research cannot be used to as the basis of a *causal relationship*, where one variable affects another. Qualitative research often has the aim of *description* and researchers may follow-up with examinations of why the observations exist and what the implications of the findings are.

The idea behind this type of research is to study frequencies, averages, and other statistical calculations. Although this research is highly accurate, it does not gather the causes behind a situation. Descriptive research is mainly done when a researcher wants to gain a better understanding of a topic. That is, analysis of the past as opposed to the future. Descriptive research is the exploration of the existing phenomena.

Although some people dismiss descriptive research as '*mere description*', good description is fundamental to the research enterprise and it has added immeasurably to our knowledge of the shape and nature of our society. Descriptive research encompasses much government sponsored research including the population census, the collection of a wide range of social indicators and economic information such as household expenditure patterns, time use studies, employment and crime statistics and the like.

Good description provokes the '*why*' questions of explanatory research. If we detect greater social polarization over the last 20 years (i.e. the rich are getting richer and the poor are getting poorer) we are forced to ask '*Why is this happening?*' But before asking '*why?*' we must be sure about the fact and dimensions of the phenomenon of increasing polarization.

2. Explanatory Research/Correlational Design

Why is it going on (explanatory research)?

Explanatory/correlational research sometimes referred to as *Causal research* is the investigation of (research into) cause-and-effect relationships. To determine causality, it is important to observe variation in the variable assumed to cause the change in the other variable(s), and then measure the changes in the other variable(s). Other confounding influences must be controlled for so they don't distort the results, either by holding them constant in the experimental creation of data, or by using statistical methods. This type of research is very complex and the researcher can never be completely certain that there are no other factors influencing the causal relationship, especially when dealing with people's attitudes and motivations. There are often much deeper psychological considerations that even the respondent may not be aware of.

Explanatory research focuses on why questions. For example, it is one thing to describe the crime rate in a country, to examine trends over time or to compare the rates in different countries. It is quite a different thing to develop explanations about why the crime rate is as high as it is, why some types of crime are increasing or why the rate is higher in some countries than in others.

The way in which researchers develop research designs is fundamentally affected by whether the research question is descriptive or explanatory. It affects what information is collected. For example, if we want to explain why some people are more likely to be apprehended and convicted of crimes we need to have hunches about why this is so. We may have many possibly incompatible hunches and will need to collect information that enables us to see which hunches work best empirically.

Answering the '*why*' questions involves developing causal explanations. *Causal explanations argue that phenomenon Y (e.g. income level) is affected by factor X (e.g. gender).* Some causal explanations will be simple while others will be more complex. For example, we might argue that there is a direct effect of

gender on income (i.e. simple gender discrimination). We might argue for a causal chain, such as that gender affects choice of field of training which in turn affects occupational options, which are linked to opportunities for promotion, which in turn affect income level. Or we could posit a more complex model involving a number of interrelated causal chains.

Objectives of Explanatory Research

Explanatory/Causal research falls under the category of conclusive research, because of its attempt to reveal a cause-and-effect relationship between two variables. Like descriptive research, this form of research attempts to prove an idea put forward by an individual or organization. However, it significantly differs on both its methods and its purpose. Where descriptive research is broad in scope, attempting to better define any opinion, attitude, or behaviour held by a particular group, causal research will have only two objectives:

1. *Understanding which variables are the cause, and which variables are the effect.* For example, let's say a city council wanted to reduce car accidents on their streets. They might find through preliminary descriptive and exploratory research that both accidents and road rage have been steadily increasing over the past 5 years. Instead of automatically assuming that road rage is the cause of these accidents, it would be important to measure whether the opposite could be true. Maybe road rage increases in light of more accidents due to lane closures and increased traffic. It could also be the case of the old adage 'correlation does not guarantee causation.' Maybe both are increasing due to another reason like construction, lack of proper traffic controls, or an influx of new drivers.
2. *Determining the nature of the relationship between the causal variables and the effect predicted.* Continuing with our example, let's say the city council proved that road rage had an increasing effect on the number of car accidents in the area. The causal research could be used for two things. First measuring the significance of the effect, like quantifying the percentage increase in accidents that can be contributed by road rage. Second, observing how the relationship between the variables works (ie: enraged drivers are prone to accelerating dangerously or taking more risks, resulting in more accidents).

The following are examples of research objectives for explanatory/causal research design:

- a. To assess the effect of foreign direct investment on the levels of economic growth in Kenya
- b. To analyze the relationship between work environment conditions and employee productivity in public universities in Kenya

In business and social sciences, researchers usually use a combination of both descriptive and explanatory research designs; referred to as "*Descripto-explanatory*" research design. This involves a combination of both descriptive and explanatory research designs. According to Saunders *et al.* (2009), in business and social sciences, it is usually common that research begins with description of variable characteristics before explanation of the causal relationship between variables. In such studies, description is thought of as a means to an end rather than an end in itself. This means that if your research utilizes description, it is likely to be a precursor to explanation.

3. Confirmatory Research Design

Confirmatory research tests *a priori* hypotheses - outcome predictions that are made before the measurement phase begins. Such *a priori* hypotheses are usually derived from a theory or the results of previous studies. The advantage of confirmatory research is that the result is more meaningful, in the sense that it is much harder to claim that a certain result is generalizable beyond the data set. The reason for this is that in confirmatory research, one ideally strives to reduce the probability of falsely reporting a coincidental result as meaningful. This probability is known as α -level or the probability of a type I error.

4. Exploratory Research Design

Exploratory Research is research conducted for a problem that has not been studied more clearly, establishes priorities, develops operational definitions and improve the final research design. Exploratory research helps determine the best research design, data-collection method and selection of subjects. It should draw definitive conclusions only with extreme caution. Given its fundamental nature, exploratory research often concludes that a perceived problem does not actually exist.

Exploratory research often relies on techniques such as:

- Secondary research - such as reviewing available literature and/or data
- Informal qualitative approaches, such as discussions with consumers, employees, management or competitors
- Formal qualitative research through in-depth interviews, focus groups, projective methods, case studies or pilot studies

When research aims to gain familiarity with a phenomenon or to acquire new insight into it in order to formulate a more precise problem or to develop a hypothesis, exploratory studies (also known as formulative research) come in handy. If the theory happens to be too general or too specific, a hypothesis cannot be formulated. Therefore, a need for exploratory research is felt to gain experience that may help in formulating a relevant hypothesis for more definite investigation. The results of exploratory research are not usually useful for decision-making by themselves, but they can provide significant insight into a given situation. Although the results of qualitative research can give some indication as to the "why", "how" and "when" something occurs, they cannot reveal "how often" or "how many".

Research Variables

A variable represents any characteristic, number, or quantity that can be measured or quantified. The term encompasses anything that can vary or change, ranging from simple concepts like age and height to more complex ones like satisfaction levels or economic status. Variables are essential in research as they are the foundational elements that researchers manipulate, measure, or control to gain insights into relationships, causes, and effects within their studies. They enable the framing of research questions, the formulation of hypotheses, and the interpretation of results.

Research variables are characteristics, quantities, or phenomena that can be measured or changed and are examined in a study to understand relationships, causes, and effects. Key types include independent variables, which are manipulated by the researcher (the cause), and dependent variables, which are measured to see the effect of the independent variable (the effect). Other important variables include control variables, kept constant, moderator variables, which change the relationship between other variables, and mediating variables, which explain the relationship between two other variables.

Another important distinction having to do with the term 'variable' is the distinction between an *independent* and *dependent* variable. This distinction is particularly relevant when you are investigating cause-effect relationships. In fact, *the independent variable is what you (or nature) manipulate*; a treatment or program or cause. The *dependent variable* is what is affected by the *independent variable*. For example, if you are studying the effects of a new educational program on student achievement, the program is the independent variable and your measures of achievement are the dependent ones.

1. *Independent Variable*; The variable that the researcher manipulates or changes to observe its effect on the dependent variable.
 - Example: In a study on the effect of temperature on test scores, the room temperature is the independent variable.
2. *Dependent Variable*; The variable that is measured to see if it is affected by the independent variable.
 - Example: In the same study, test scores are the dependent variable because they depend on the room's temperature.
3. *Moderator Variable*; A variable that changes the strength or direction of the relationship between the independent and dependent variables.
 - Example: Personality type could moderate the relationship between a study method and test scores; perhaps introverts benefit more from one method than extroverts.
4. *Mediating Variable*; A variable that explains the relationship between the independent and dependent variables.
 - Example: In a study on how education affects income, job skills could be a mediating variable, as they explain why education leads to higher income.

Operationalization of the Research Variables

In research design, especially in psychology, social sciences, life sciences, and physics, *operationalization* is a process of defining the measurement of a phenomenon that is not directly measurable, though its existence is indicated by other phenomena. Operationalization is thus the process of defining a fuzzy concept so as to make it clearly distinguishable, measurable, and understandable in terms of empirical observations. In a wider sense, it refers to the process of specifying the extension of a concept - describing what is and is not an instance of that concept.

Examples:

1. In Entrepreneurship, the concept Innovation may be operationalized into new products, new processes, improved products etc. while Growth may be operationalized as new markets or growth in market share, rise in sales, business expansion, rise in profits etc.
2. In HRM, the phenomenon of Employee Performance might be operationalized by one or more indicators like completing tasks on time, quality of work outputs, meeting targets etc.

Research Population and Sample

In research, a population is the entire group of individuals, items, or events with shared characteristics that a study aims to generalize its findings to. It's the complete set from which a sample is drawn, which is the smaller, manageable group from which data is actually collected. Researchers clearly define the population's characteristics to ensure their findings are accurate and applicable to the broader group of interest.

It is an *aggregate or totality of all the objects, subjects or members that conform to a set of specifications*. A population is the total of all the individuals who have certain characteristics and are of interest to a researcher. Community college students, race car drivers, teachers, college-level athletes, and disabled war veterans can all be considered populations.

A research population is generally a large collection of individuals or objects that is the main focus of a scientific query. It is for the benefit of the population that researches are done. However, due to the large sizes of populations, researchers often cannot test every individual in the population because it is too expensive and time-consuming. This is the reason why researchers rely on sampling techniques. A research population is also known as a well-defined collection of individuals or objects known to have similar characteristics.

Key Aspects of a Research Population

- *The Whole Group*; The population includes every single element that matches the study's criteria, whether that's people, organizations, or even objects.
- *Shared Characteristics*; Elements within the population share specific traits that are relevant to the research question. For example, the population might be "adults over 65 in a particular city" or "all diagnosed cases of a specific disease in a hospital".
- *Target Population*; This is the specific subgroup of the larger population that the research is primarily concerned with and to which the results will be generalized.
- *Study Population*; A subset of the target population that is actually accessible and available for the study.
- *Sample*; The subset of the study population from which data is collected. Researchers study the sample to make inferences about the entire population.

Simply put, a population is the entire group you want to draw conclusions about, while a sample is the smaller, specific group (subset) you collect data from to make those inferences. The sample is a subset of the population, and its representativeness is crucial for accurately estimating the characteristics of the whole group.

The reason why samples are important is that within many models of scientific research, it is impossible (from both a strategic and a resource perspective) to study all the members of a population for a research

project It just costs too much and takes too much time. Instead, selected few participants (who make up the sample) are chosen to ensure that the sample is representative of the population. And, if this is the case, then the results from the sample can be inferred to the population, which is exactly the purpose of inferential statistics, using information on a smaller group of participants to infer to the group of all participants.

There are many types of samples; *A random sample*, *A stratified sample*, and *A convenience sample*. They all have the goal of accurately creating a smaller subset from the larger set of general participants such that the smaller subset is representative of the larger set.

- i. *Random Sample*; A random sample is a subset of individuals chosen from a larger population where every member has an equal chance of being selected. This method is used in statistics to gather data from a smaller group to make inferences about the entire population, reducing bias and helping researchers generalize findings.
- ii. *Stratified Sample*; A stratified sample first divides the population into subgroups (strata) based on shared characteristics, and then randomly samples from each subgroup to ensure representation.
- iii. *Convenience Sample*; A convenience sample is a non-probability sampling method where participants are chosen based on their easy accessibility and proximity to the researcher. This method is quick and inexpensive, but it carries a high risk of bias and may not produce results that are representative of the larger population. Examples include surveying people in a specific shopping mall or asking students in your own class.

Census versus Sample

A census collects data from every single member of a population, providing complete data but requiring significant time and cost, while a sample collects data from a smaller, representative subset of the population, offering a faster, more cost-effective approach with a slightly lower level of accuracy. The choice between a census and a sample depends on factors like available resources, time constraints, and the specific goals of the research. Census studies occur when the size of the population is small and manageable, hence no need for sampling.

Sampling Techniques

In statistics, sampling is the process of selecting a subset of data from a larger dataset. There are two main types of sampling: probability sampling and non-probability sampling. The main difference between the two types of sampling is how the sample is selected from the population.

a) Probability Sampling

Probability sampling is a type of sampling where every member of the population has a known and equal chance of being selected for the sample. This means that the sample is chosen randomly from the population, using a random number generator or other methods to ensure that each member of the population has an equal chance of being included in the sample. Probability sampling is considered the most reliable and unbiased method because it ensures that the sample is representative of the population and reduces the potential for bias.

b) Non-probability Sampling

Non-probability sampling is a type in which the sample members are not randomly selected from the population. In non-probability sampling, the sample may be selected based on convenience, availability, or other factors rather than random selection. Non-probability sampling is generally considered less reliable and less unbiased than probability sampling because it is not guaranteed to be representative of the population.

In this type of population sampling, members of the population do not have equal chance of being selected. Due to this, it is not safe to assume that the sample fully represents the target population.

Types of Probability Sampling

Several different sampling methods can be used for probability sampling. Some common sampling methods for probability sampling include:

- i. *Simple random sampling*: This is the most basic method of probability sampling, where every member of the population has an equal chance of being selected for the sample. This is typically

done using a random number generator, lottery or other methods to select a subset of the population randomly.

- ii. *Systematic random sampling*: This is a probability sampling method where the sample is selected using a systematic, random approach. This method involves selecting a random starting point in the population and then selecting every kth member of the population (for example, checking every 6th piece produced by the machine) to be included in the sample. For instance, if the population contains 1000 members, and the sample size is 100, the researcher could select a random number between 1 and 10 and then select every 10th member of the population starting from that point. This method ensures that every member of the population has an equal chance of being selected for the sample and can be more efficient than simple random sampling.
- iii. *Stratified sampling*: This method involves dividing the population into different subgroups, or strata, based on certain characteristics and then randomly selecting a sample from each stratum. This ensures that the sample is representative of the various subgroups in the population. For example, if ten people are drawn to represent a country, five of them are male and five are female, to avoid gender bias.
- iv. *Cluster sampling*: This method involves dividing the population into different clusters and then randomly selecting a sample of clusters. All members of the selected clusters are then included in the sample. This method is often used when it is impractical to sample the entire population and can be more efficient than simple random sampling. Sampling is often clustered by geography or by time periods. For example, survey all customers visiting particular stores on particular days.

Types of Non-probability Sampling

1. Purposive sampling

This is a non-representative subset of some larger population, and is constructed to serve a very specific need or purpose. A researcher may have a specific group in mind, such as high level business executives. It may not be possible to specify the population -- they would not all be known, and access will be difficult. The researcher will attempt to zero in on the target group, interviewing whomever is available.

A subset of purposive sampling is:

- a) *Snowball Sampling* - a non-probability sampling technique that is used by researchers to identify potential subjects in studies where subjects are hard to locate. Researchers use this sampling method if the sample for the study is very rare or is limited to a very small subgroup of the population. This type of sampling technique works like chain referral. After observing the initial subject, the researcher asks for assistance from the subject to help identify people with a similar trait of interest.
- b) *Convenience sampling*: This method involves selecting a sample based on convenience or availability. For example, a researcher may choose a sample of participants from a nearby community or a convenient location rather than selecting a random sample from the population. It's usually done when you want to get as many responses as possible quickly. Convenience samples are typically biased toward those who agree with your research question. For example, surveying friends and family members.

2. Judgement sampling

This is a method of non-probability sampling where the sample is selected based on the judgement or expertise of the researcher. In judgement sampling, the researcher uses their knowledge and experience to select a sample that is believed to represent the population. This method is often used when it is difficult or impossible to sample the population randomly or when the researcher has expertise in a particular field that allows them to select a representative sample. However, judgment sampling is generally considered less reliable and less unbiased than probability sampling because it is not guaranteed to be representative of the population. For example, an auditor selects a sample based on their concerns in the earlier audit.

3. Disproportional Sampling

This is a non-probability sampling technique used to address the difficulty researchers' encounter with stratified samples of unequal sizes.

Example: Suppose, for example, a researcher desires to conduct a survey of all the students in a given university with 10,000 students; 8,000 females and 2,000 males. His desired sample size is only 1,000. Since the 1,000 subjects needed for the survey is 10% of the entire population, sampling proportion suggests that 8/10 be female and 2/10 be male. This would result in a sample composed of 800 females and 200 males. In this case, the relatively small number of males in the sample probably would not provide adequate representation for drawing conclusions from the said survey.

Sampling Frame vs Sampling Unit

In statistics, a *sampling frame* is the source material or device from which a sample is drawn. It is a list of all those within a population who can be sampled, and may include individuals, households or institutions. It is the list of all available members of a target population from which a sample will be drawn, while a *sampling unit* is the individual element within that list that is selected for the sample to collect data from. For instance, if a study wants to sample households in a city, the sampling frame is the complete list of all households, and each individual household on that list is a sampling unit.

The most straightforward type of frame is a *list of elements of the population* (preferably the entire population) with appropriate contact information. For example, in an opinion poll, possible sampling frames include an electoral register or a telephone directory. Other sampling frames can include *employment records, school class lists, patient files in a hospital, organizations listed in a thematic database*, and so on.

The sampling unit is a component of the sampling frame. You first define the sampling frame (the entire list) and then you select the sampling units (individual items on the list) from the frame to form your sample.

Data Types and Sources

Data can be categorized by **type**, including quantitative (numerical) and qualitative (descriptive) data, and by **source**, into primary (original) or secondary (previously collected) data, and internal (within an organization) or external (from outside) sources. These categories help researchers and analysts select appropriate methods for data collection and analysis to address specific goals.

Data sources are the places or origins from which data is collected for analysis, reporting, or other uses. Researchers need to consider the sources on which to base and confirm their research and findings. They have a choice between primary data and secondary sources and the use of both, which is termed triangulation, or dual methodology.

Data Types

- a) *Quantitative Data*; Numerical data that can be measured, calculated, and expressed in numbers, such as statistics, percentages, or time.
- b) *Qualitative Data*; non-numerical, descriptive data that provides context and insight into behaviors and actions, such as text from interviews, descriptions, or observations.

Data Sources

1. Primary Data

When someone refers to "primary data" they are referring to data collected by the researcher himself/herself. This is data that has never been gathered before, whether in a particular way, or at a certain period of time. Researchers tend to gather this type of data when what they want cannot be found from outside sources. You can tailor your data questions and collection to fit the need of your research questions. This can be an extremely costly task and, if associated with a college or institute, requires permission and authorization to collect such data.

Issues of consent and confidentiality are of extreme importance. Primary data actually follows behind secondary data because you should use current information and data before collecting more so you can be informed about what has already been discovered on a particular research topic.

Primary data is the data collected by the researcher themselves. They include; a document or record containing first-hand information or original data on a topic. A work created at the time of an event or by a person who directly experienced an event. Such data can be obtained through *interview, observation, questionnaires* etc.

2. **Secondary Data**

This type of data typically comes from other studies done by other institutions or organizations. There is no less validity with secondary data, but you should be well informed about how it was collected. Secondary sources are data that already exists - *Previous research, Official statistics, Mass media products, Diaries, Letters, Government reports, Web information, Historical data and information.*

Any published or unpublished work that is one step removed from the original source, usually describing, summarizing, analyzing, evaluating, derived from, or based on primary source materials. A source that is one step removed from the original event or experience and that provides criticism or interpretation of a primary source. Some examples include *textbooks, review articles, biographies, historical films, music and art, articles about people and events from the past.*

Data Collection Methods and Instruments

Data collection is the process of gathering and measuring information on targeted variables in an established systematic fashion, which then enables one to answer relevant questions and evaluate outcomes. Data collection is a component of research in all fields of study including physical and social sciences, humanities, and business. While methods vary by discipline, the emphasis on ensuring accurate and honest collection remains the same. The goal for all data collection is to capture quality evidence that allows analysis to lead to the formulation of convincing and credible answers to the questions that have been posed.

The quality of research depends to a large extent on the quality of data collection tools. Interviewing and administering questionnaire are probably the most commonly used research techniques. This makes an Interview Guide and Questionnaire the most commonly used instruments in research. Therefore, designing good 'questioning tools' forms an important and time-consuming phase in the development of a research proposal. Once a decision has been made to use the two instruments, the following questions should be considered before designing them:

- a) What exactly do we want to know, according to the objectives and variables we have identified?
- b) Is questioning the right technique to obtain answers or do we need additional techniques such as *observations* or *analysis of records*?
- c) Of whom will we ask questions and what techniques will we use? Do we understand the topic sufficiently to design a questionnaire or do we need some loosely structured interviews with *key informants*?
- d) Are our informants mainly literate or illiterate?

How large is the sample that will be interviewed? Studies with many respondents often use shorter, highly structured questionnaires while smaller studies allow more flexibility and may use interview guides or questionnaires with open-ended questions. To supplement interviewing, one may also need to develop an *Observational Guide*. The observational guide details the items to be observed and what is observed recorded accordingly.

Observational guides are particularly important to help confirm responses practically especially in studies on assessing impact or changes in behaviour and practice. The design and administration of both the interview guide and questionnaire take the same approach and precision, whereas that of the observational guide is dictated by the variables or issues of concern and therefore, what to look for in order to confirm the real situation.

Regardless of the field of study or preference for defining data (quantitative or qualitative), accurate data collection is essential to maintaining the integrity of research. Both the selection of appropriate data collection instruments (existing, modified, or newly developed) and clearly delineated instructions for their

correct use reduce the likelihood of errors occurring. A formal data collection process is necessary as it ensures that the data gathered are both defined and accurate and that subsequent decisions based on arguments embodied in the findings are valid. The process provides both a baseline from which to measure and in certain cases an indication of what to improve.

Consequences from improperly collected data include:

- Inability to answer research questions accurately;
- Inability to repeat and validate the study.

Distorted findings result in wasted resources and can mislead other researchers into pursuing fruitless avenues of investigation; it may also compromise decisions, for example for public policy, which may cause disproportionate harm.

A number of research methods and instruments are available for consideration in collecting data in a research project. The key and most common ones in business and social sciences are interview guide, questionnaire, and observational guide. The decision as to which instrument is most appropriate will be informed by the data requirements and its sources.

1. Interview and Interview Guide

In social science, *interviews* are a method of data collection that involves two or more people exchanging information through a series of questions and answers. The questions are designed by a researcher to elicit information from interview participants on a specific topic or set of topics. These topics are informed by the author's research questions. Interviews typically involve an in-person meeting between two people (an interviewer and an interviewee), but interviews need not be limited to two people, nor must they occur in-person.

You may be wondering when you should choose interviews as your data collection method. Interviews are an excellent way to gather detailed information. They also have an advantage over surveys, as they can be adapted as you learn more information. Recall that survey data collection methods do not allow researchers to change the questions that are administered, even if a participant's response sparks some follow-up question in your mind. All participants must be asked the same questions in the same manner.

The questions you decided to put on your survey during the design stage determine what data you get. In an interview, however, you can follow up on new and unexpected topics that emerge during the conversation. Trusting in emergence and learning from your participants are hallmarks of qualitative research. In this way, interviews are a useful method to employ when you want to know the story behind the responses you might receive in a written survey.

Types of Interviews

1. Structured Interviews

Structured interviews are defined as research tools that could be more flexible in their operations and allows more or no scope of prompting the participants to obtain and analyze results. It is thus also known as a standardized interview and is significantly quantitative in its approach.

Questions in this interview are pre-decided according to the required detail of information. This can be used in a focus group interview and an in-person interview. These interviews are excessively used in survey research with the intention of maintaining uniformity throughout all the interview sessions.

They can be closed-ended and open-ended – according to the type of target population. Closed-ended questions can be included to understand user preferences from a collection of answer options. In contrast, open-ended ones can be included to gain details about a particular section in the interview.

Examples of a structured interview questions

Here's an example of a structured question for a job interview for a customer service job:

- a) Can you talk about what it was like to work in customer service?

- b) How do you deal with an angry or upset customer?
- c) How do you ensure that the information you give customers is correct?
- d) Tell us about when you went out of your way to help a customer.
- e) How do you handle a lot of customers or tasks at once?
- f) Can you talk about how you've used software or tools for customer service?
- g) How do you set priorities and use your time well while giving good customer service?
- h) Can you tell us about when you had to get a customer to calm down?
- i) How do you deal with a customer who wants something that goes against your company's rules?
- j) Tell me about a time when you had to deal with a hard customer or coworker.

2. Semi-Structured Types of Interviews

Semi-structured interviews offer a considerable amount of leeway to the researcher to probe the respondents, along with maintaining a basic interview structure. Even if it is a guided conversation between researchers and interviewees – appreciable flexibility is offered to the researchers. A researcher can be assured that multiple interview rounds will not be required in the presence of structure in this type of research interview.

Keeping the structure in mind, the researcher can follow any idea or take creative advantage of the entire interview. Additional respondent probing is always necessary to garner information for a research study. The best application of semi-structured interviews is when the researcher doesn't have time to conduct research and requires detailed information about the topic.

Example of a semi-structured interview question

Here's an example of a semi-structured marketing job interviews question.

- i. Can you tell us about the marketing work you've done?
- ii. What do you think are the most important parts of a marketing campaign that works?
- iii. Tell me about a campaign you worked on that you're very proud of.
- iv. How do you do research on the market and look at data to help you make marketing decisions?
- v. Can you tell us about a time when you had to change your marketing plan because of something that didn't go as planned?
- vi. How do you figure out if a marketing campaign worked?
- vii. Can you talk about how you've used social media to market?
- viii. How do you ensure your marketing message gets through to the people you want to hear it?
- ix. Can you tell us about a time when you had to run a marketing campaign on a small budget?
- x. How do you keep up with changes and trends in marketing?

3. Unstructured Interviews

Also called in-depth interviews, unstructured interviews are usually described as conversations held with a purpose in mind – to gather data about the research study. These interviews have the least number of questions as they lean more towards a normal conversation but with an underlying subject.

The main objective of most researchers using unstructured interviews is to build a bond with the respondents, due to which there is a high chance that the respondents will be 100% truthful with their answers. There are no guidelines for the researchers to follow. So, they can approach the participants ethically to gain as much information as possible about their research topic.

Since there are no guidelines for these interviews, a researcher is expected to keep their approach in check so that the respondents do not sway away from the main research motive. For a researcher to obtain the desired outcome, he/she must keep the following factors in mind:

- a) The intent of the interview.
- b) The interview should primarily take into consideration the participant's interests and skills.
- c) All the conversations should be conducted within the permissible limits of research, and the researcher should try and stick by these limits.
- d) The researcher's skills and knowledge should match the interview's purpose.
- e) Researchers should understand the dos and don'ts of it.

Examples of an unstructured interview questions

Here's an example of a question asked in an unstructured interview:

- a) Can you tell me about when you had to deal with something hard and how you did it?
- b) What are some of the things you're most proud of, and what did you learn from them?
- c) How do you deal with ambiguity or not knowing what to do at work?
- d) Can you describe how you lead and how you get your team going?
- e) Tell me about a time when you had to take a chance and how it turned out.
- f) What do you think are the most important qualities for success in this role?
- g) How do you deal with setbacks or failures, and what do you learn from them?
- h) Can you tell me about a time when you had to solve a problem by thinking outside the box?
- i) What do you think makes you different from the other people who want this job?
- j) Can you tell me about a time when you had to make a hard choice and how you made that choice?

As noted earlier an interview guide is one of the major research instruments in social sciences. An interview is a purposeful discussion between an interviewer and a respondent in which the interviewer asks questions and records responses. The use of interviews can help gather valid and reliable data relevant to the research questions and objectives. From this, an interview guide can be defined as a set of questions prepared to guide in getting data or responses from a respondent.

An interview may be highly formalized and structured, using standardized questions for each respondent or they may be informal and unstructured conversations. A structured interview guide has a set of questions, which the interviewer reads, and records the responses on a standardized schedule whereas in a semi-structured and non-standardized interview guide - there are a list of themes and questions to be covered which may vary from interview to interview.

What this means is that the interview may omit certain questions in particular interviews, given the specific organizational context that is encountered in relation to the research topic. The questions may also be varied depending on the flow of the conversation. In such interviews, additional questions may be required to explore the research question and objectives given the nature of events within a particular organization. Given the nature of the questions and the ensuing discussion, data is recorded by note – taking or tape recording the conversation.

Whether structured or semi-structured, an interview guide may be administered on a face-to-face basis or through telephone when it is a single respondent. There are also situations where an interview guide may be administered to a group of respondents usually referred to as *focused group interviews* – where you meet a small group of respondents to explore an aspect of research through group discussion that you facilitate.

Measures to avoid Bias in Interviews

Yielding reliable and valid data is the focus in the use of any research instrument. Some of the precautions you need to take in order yield such information include:

- i. *Prepare adequately and be ready for the interview* – be knowledgeable about the organizational or situational context in which the interview is to take place. This may help in assessing the accuracy of the information given by respondents.
- ii. *Provide enough and adequate information to the interviewee* – supply relevant information to the respondents before the interview; themes to enable them prepare and give you adequate information.
- iii. *Appear at the interview with the right dressing and at the appropriate time* – the way you dress for the interview may have an effect on your credibility in the view of the interview of the respondent and thus the response. Wear clothing that is generally acceptable for the setting within which the interview is to take place.
- iv. *Make calculated and appropriate comments when commencing the interview* – the willingness and cooperation of the respondent to give information will depend on the initial approach. The respondent may need to clarification on the exact nature of the data required and why and may in

some cases develop interest to participate in the research. The opening moves should demonstrate credibility and friendliness.

- v. *Approach the questioning with ease and in a friendly manner* – rephrase questions clearly so that respondent can easily understand them. Start with open questions then move to probing questions. In addition, avoid long and confusing questions.
- vi. *Behave decently during the interview* – avoid any non-verbal behaviour that may indicate bias. The posture and tone of voice are also issues to consider, as they are likely to inhibit the flow of the discussion. Sit slightly inclined to the respondent and adopt an open posture, avoiding folded arms to show attentiveness to your respondents.
- vii. *Listen to the respondent attentively* – do not compete to speak with the respondent! Listening involves people being on the lookout for signals and willing to spend the time needed to listen and build understanding, deliberately holding back their thoughts, which would divert or compete with others.
- viii. *Understand the responses fully* – summarize each explanation provided by the respondent to allow him evaluate the adequacy of the interpretation and correct where necessary.
- ix. *Record information as accurately as given* – record the information as given. It is always in order to ask for clarification in responses that you do not understand than record what you think was the response!

4. Survey Research and Questionnaire

Survey Research is a quantitative research method used for collecting data from a set of respondents. It has been perhaps one of the most used methodologies in the industry for several years due to the multiple benefits and advantages that it has when collecting and analyzing data.

Survey Research is defined as the process of conducting research using surveys that researchers send to survey respondents. The data collected from surveys is then statistically analyzed to draw meaningful research conclusions.

Survey Research Methods

Survey research methods can be derived based on two critical factors: Survey research tool and time involved in conducting research. There are three main survey research methods, divided based on the medium of conducting survey research:

- i. *Online/Email:* Online survey research is one of the most popular survey research methods today. The survey cost involved in online survey research is extremely minimal, and the responses gathered are highly accurate.
- ii. *Phone:* Survey research conducted over the telephone can be useful in collecting data from a more extensive section of the target population. There are chances that the money invested in phone surveys will be higher than other mediums, and the time required will be higher.
- iii. *Face-to-face:* Researchers conduct face-to-face in-depth interviews in situations where there is a complicated problem to solve. The response rate for this method is the highest, but it can be costly.

How to Create a Survey with a Good Design

As explained before, a survey usually has its beginnings when a person, company, or organization faces a need for information, and there is no existing data that is sufficient. Take into account the following recommendations:

- a) *Define objective:* The survey would have no meaning if the aim and the result were unplanned before deploying it. The survey method and plan should include actionable milestones and the sample planned for research. Appropriate distribution methods for these samples also have to be put in place right at the outset.
- b) *The number of questions:* The number of questions used in a market research study depends on the research's end objective. It is essential to avoid redundant queries in every way possible. The length of the questionnaire has to be dictated only by the core data metrics that have to be collected.
- c) *Simple language:* One factor that can cause a high survey dropout rate is if the respondent finds the language difficult to understand. Therefore, it is imperative to use easily understandable text in the survey.

- d) *Question types*: There are several types of questions that can go into a survey. It is essential to use the question types that offer the most value to the research while being the easiest to understand and answer to a respondent.
- e) *Consistent scales*: If you use rating scale questions, ensure the scales are consistent throughout the research study. Using scales from -5 to +5 in one question and -3 to +3 in another question may confuse a respondent.
- f) *Survey Logic*: Logic is one of the most critical aspects of the survey design. If the logic is flawed, respondents will not be able to continue further or the desired way. Logic must be applied and tested to ensure that only the next logical question appears when selecting an option.

A survey questionnaire is an instrument used for systematic collection of information from a selected group of people by asking a series of questions. A survey can be used to collect various types of information – beliefs, people’s behaviour, job performance, knowledge, preferences, attitudes and feelings among others. Wai - Ching of the University of East Anglia, Norwich defines a survey as a *method of collecting information from a sample of the population or sometimes the organizations of interest*. Survey as an instrument can be used when it is faster, easier or less expensive and when the information required does not already exist in some form.

Types of Questions in Research

There are two main types of questions in research; structured and unstructured.

1) Structured Questions

Structured questions in research are closed-ended, predetermined questions with fixed response options like multiple-choice or yes/no, designed to collect quantitative data efficiently and allow for easy comparison across participants. They are primarily used in large-scale surveys and structured interviews to gather quantifiable information and ensure uniformity, though they can limit depth and prevent respondents from providing nuanced answers.

Key characteristics of structured questions include:

- a) *Closed-ended*: Participants choose from a predefined set of answers.
- b) *Quantitative data*: The format is ideal for generating numerical data that can be statistically analyzed.
- c) *Standardized*: Questions are asked in the same order to all participants, minimizing bias and ensuring consistency.
- d) *Examples*: Multiple-choice, yes/no, and Likert scale questions are common forms.

Common examples of structured questions include:

- i. *Multiple-choice*: These give a list of options to pick from. For example, "What's your favourite ice cream flavour? a) Chocolate, b) Vanilla, c) Strawberry, d) Other". The multiple choice question allows respondents to choose from one or more of these answers, depending on what you decide.
- ii. *Likert scale*: This type of question asks respondents how much they agree or disagree with something. It usually goes from "Strongly Disagree" to "Strongly Agree". For example, "I enjoy reading books: Strongly Disagree / Disagree / Neutral / Agree / Strongly Agree".
- iii. *Dichotomous*: These are yes or no questions. Simple and straight to the point! For instance, "Do you own a car? Yes / No".
- iv. *Single-choice*: In this case, there is only one correct answer for the respondent. When asked, "Where were you born?" they will have to reply with one specific country or city (i.e., Nairobi).

When to use Structured Questions

Structured questions excel at:

- a. *Quantitative research*: If you want numerical data to quantify outcomes, metrics, or opinions - use structured questions. For example, customer satisfaction rating scales.
- b. *Standardised data collection*: If you need to easily compare/contrast responses across a population, use consistent structured questions. For example, brand awareness metrics.

- c. *Large respondent samples*: Structured questions make it feasible to gather and process data from thousands of respondents.

Structured questions are popular in many fields. Market research often uses them to get quick feedback on products. Political polls use them to gauge public opinion quickly. Customer satisfaction surveys often use rating scales to measure how happy clients are with a service.

2) Unstructured Questions

Unstructured questions are the opposite of what we just defined. They are open-ended, allowing respondents to answer however they want in free-form writing. Common examples include:

- a. *Open-ended questions*: These let respondents write whatever they want. Like "What do you think about our new product?"
- b. *Probing questions*: These often come after a structured question to get more info. For example, "Why did you choose that rating?"
- c. *Hypothetical questions*: Ask what would you do in a made-up situation. Such as "If you were in charge for a day, what would you change?"
- d. *Essay-style responses*: These are for longer, more detailed answers. Like "Describe your ideal vacation and why you'd enjoy it."

Unstructured questions lead to qualitative, detailed responses with more context. But they make standardizing and analysis more difficult.

When to use Unstructured Questions

Structured questions might be more common, but unstructured ones shine in certain situations. They're great when you need:

- i. *Exploratory research*: If you're still developing a concept and want detailed, contextual insights on user perspectives, use open-ended questions. This is common in product development research.
- ii. *Capturing detailed personal experiences*: Let respondents tell their full story by using unstructured questions when you want rich, qualitative details on a personal experience. For example, customer service feedback.
- iii. *Small to medium-sized respondent samples*: When you have time to read and analyse detailed responses.

Unstructured questions are often used in fields like psychology and sociology. They're great for understanding complex human experiences. Product development teams might use them to get detailed feedback on new ideas. Journalists often use open-ended questions to get the full story in interviews.

Combining Structured and Unstructured Questions

Rather than choosing one or the other, many surveys combine structured and unstructured questions to get the best of both worlds. Using both types gives you the best of both worlds. You get easy-to-analyse data from structured questions and detailed insights from unstructured ones. But be careful! Too many questions, especially open-ended ones, can make your survey too long.

Useful tips for using both types include:

- a) Start with structured questions to get basic info
- b) Use unstructured questions to dig deeper into interesting topics
- c) Keep your survey balanced - don't overload it with too many open-ended questions
- d) Put the most important questions first in case people don't finish the survey
- e) Only add open-ended questions when additional detail is truly needed

5. Observation and Observational Guide

Observation, as the name implies, is a way of collecting data through observing. This data collection method is classified as a participatory study, because the researcher has to immerse herself in the setting where her respondents are, while taking notes and/or recording. Observation data collection method may involve watching, listening, reading, touching, and recording behavior and characteristics of phenomena.

Observation as a data collection method can be structured or unstructured. In structured or systematic observation, data collection is conducted using specific variables and according to a pre-defined schedule.

Unstructured observation, on the other hand, is conducted in an open and free manner in a sense that there would be no pre-determined variables or objectives.

Types of Observation

There are several types of observational methods used across many social science disciplines. They're each suited to different research purposes, and have different strengths and limitations.

It's important to note that observational methods can be *covert* or *overt*.

- In *covert research*, the research participants don't know who the researcher is, or that there's even a researcher there at all.
- In *overt research*, the research participants are all aware of the researcher's presence and their role as an observer.

a) Participant Observation

In participant observation, the researcher integrates themselves into a group to study their way of life, their culture, and how they structure their community. This technique is commonly used in ethnography.

Ethnography is the study of the way of life of a group or community.

The fact that researchers have to be integrated into the group's way of life means that they need to find a way to be let into the community. However, many communities don't want to be studied. So, the researcher can either earn the trust of certain members and seek permission to study their way of life (overt observation), or the researcher can pretend to become a member of the group to gain access to information (covert observation).

While conducting participant observation, the researcher should focus on capturing an accurate and authentic account of the community's way of life. This means that the researcher has to avoid influencing the behaviour of anyone in the group. Where simply observing the crowd isn't enough, the researcher might need to ask some questions. If they're conducting covert research, they might enlist an informant. The informant *will* be aware of the researcher's presence and can answer questions that are not addressed by observation alone.

Taking notes is more difficult when they're acting covertly. It's common for researchers to pop into the bathroom to make a quick note of something important, or to summarize their daily observations every evening. Where the researcher's presence is known, it's relatively simple for them to take notes, because they don't need to hide the fact that they're conducting research.

b) Non-participant Observation

In **non-participant observation**, the researcher studies their subjects from the sidelines - they don't participate or integrate themselves into the lives of the group they are studying. Non-participant observation can be either *structured* or *unstructured*. Structured non-participant observation involves some sort of observation schedule. Before they begin their observation, researchers make a list of behaviours that they expect to see. They then use this list to tick off what they see. Unstructured observation is the opposite of this - it involves the researcher freely noting down whatever they see.

Moreover, non-participant research can be overt. This is where the subjects are aware that they are being studied (like the headteacher sitting at the back of a class for one day each term). Or, the research can be covert, where the researcher's presence is a little more unassuming - the subjects don't know that they're being researched. For example, a researcher might be disguised as another customer in a shop, or use a one-way mirror.

As strange as it might sound, it's important for researchers to not only take note of what the subjects *are* doing but also what they *aren't* doing. For example, if a researcher was examining customer behaviour in a retail store, they might observe that people ask shopkeepers for assistance in some situations, but not others. What are those particular situations? What do customers do when they're uncomfortable asking for help?

Example of Observational Research

The researcher observes customers buying products in a mall. Assuming the product is soap, the researcher will observe how long the customer takes to decide whether he likes the packaging or comes to the mall with his decision already made based on advertisements. If the customer takes their time making a decision, the researcher will conclude that packaging and information on the package affect purchase behavior. If a customer makes a quick decision, the decision is likely predetermined. As a result, the researcher will recommend more and better advertisements in this case.

Designing the Research Instrument

Designing a good questionnaire takes several drafts. In the first draft we concentrate on the *content* and in the second we critically look at the *formulation and sequencing of the questions*. This is followed by scrutinizing the format of the questionnaire and test; running to check whether the questionnaire gives us the information we require and whether the interviews as well as respondents feel at ease with it. Once this is done, the questionnaire may need some adaptation before we can use for actual data collection.

It should always be kept in mind that the survey should be as brief as possible and should create as little frustration as possible to increase the likelihood that it will be completed and returned. The purpose is to help the people you are surveying give you the information you need in a form that is useful. To achieve, it is important to observe the following in designing a questionnaire:

- ✓ Keep it simple
- ✓ Make the questions specific
- ✓ Start with easier questions and move on to difficult ones; the first questions should hook the reader into answering the survey questions
- ✓ Ask questions in a logical order
- ✓ Include instructions for answering all questions included on the survey
- ✓ Keep each question brief

If the questionnaire is completed by the respondent and not by an interviewer or survey staff member, remember to design a questionnaire that is graphically pleasing and easy to read. Remember to pre-test the questionnaire - before taking the survey to the printer, ask a few people-such as regular customers, colleagues, friends or employees-to complete the survey.

- i. Ask them for feedback on the survey's style, simplicity and their perception of its purpose.
- ii. Mix the form of the questions – use scales, rankings, open-ended questions and closed-ended questions for different sections of the questionnaire. The "form" or way a question is asked may influence the answer given. Basically, there are two question forms: closed-end questions and open-end questions.
- iii. Provide clear and sufficient directions – reasons of the survey, whether responses are to be anonymous or confidential, how the survey will be completed.

Generally, the questions or statements in the survey form should;

- a) Be clearly written, short and easy to read, no jargon or acronyms unfamiliar to the respondents;
- b) Be concise, to the point;
- c) Be specific, focus on one idea at a time;
- d) Be explicit – don't force people to guess what is being asked. Be clear on what kind of information you want and
- e) Use precise wording to avoid double meaning. The point is that a properly designed survey form should have questions that are easy to understand and respond to!

Steps in Questionnaire / Interview Guide Design

1. *Content* - Decide what questions will be needed to measure. When developing questionnaires, you should consider the variables you have chosen and if necessary, add, drop or change some. You may even have to change some of your objectives at this stage.
2. *Formulating questions* - Formulate one or more questions that will provide the information needed for each variable. Take care that questions are specific and precise enough so that different

respondents do not interpret them differently. Other important precautions are that you should avoid ‘*leading questions*’ – questions that suggest a certain answer, avoiding words with double meanings and sensitive questions.

3. *Sequencing the questions* - Design the questionnaire or interview guide in such a way that it is ‘informant friendly’. This can be achieved by sequencing the questions logically to allow as much as possible for a ‘*natural*’ conversation; at the beginning, ask a limited number of questions concerning background; age, education, marital status etc.; start with an interesting, but non-controversial question that is directly related to the subject of study – to raise the respondent’s interest and lessen suspicion; pose more sensitive questions as late as possible in the interview – questions on income, political matters, sexual behaviour etc.; use simple, everyday language and use local terminologies.
4. *Formatting the Questionnaire* - While formatting the questionnaire, the following precautions should be observed; Provide a separate, introductory page explaining the purpose of the study, requesting respondent’s consent to be interviewed and assuring confidentiality of the data obtained, Start with fairly simple questions, End with easy questions, Keep the survey as short as possible, Eliminate questions that do not meet your objectives, Make sure there are no redundant questions, Make sure to include a place in the questionnaire for a survey number, date and location of the interview and if required the name of the respondent. You may also add the name of the interviewer to facilitate quality control, provide sufficient space for answers to open-ended questions, categories such as ‘other’ and for comments on supporting answers given.
5. *Translation* - There are cases where the interview may be conducted in one or more local languages. In such situation, the questionnaire / guide should be translated in order to standardize the way questions will be asked. After having translated, you should have it retranslated into the original language by a different person and then compare the two versions for differences to help rephrasing of difficult concepts.

Pilot Testing or Pre-testing

Pre-testing questionnaire will help to further check and refine the questions and try out the survey before a wide – scale distribution to make it easy to understand and provide data. The pre-testing is done with a small group of respondents of the larger group. While doing this; explain the purpose of the survey, get comments on each question – order and format, take note of questions that are frequently unanswered, and time how long it takes for respondents to answer the questionnaire. The utility of questionnaire pre-testing is to use the information obtained from the pilot test to modify the survey.

Pilot testing covers multiple activities that allow you to evaluate the different aspects of your project ahead of time. You can also describe this as a test run or rehearsal to help you discover any hitches or gaps in your plan before you begin its execution.

You can determine the feasibility of your research design, with a pilot study before you start. This is a preliminary, small-scale “rehearsal” in which you test the methods you plan to use for your research project. You will use the results to guide the methodology of your large-scale investigation. Pilot studies should be performed for both qualitative and quantitative studies. Here, we discuss the importance of the pilot study and how it will save you time, frustration and resources.

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You never test the dept of a river with both feet” – African Proverb

Pilot studies should be routinely incorporated into research designs because they:

- i. Help define the research question

- ii. Test the proposed study design and process. This could alert you to issues which may negatively affect your project.
- iii. Educate yourself on different techniques related to your study.
- iv. Test the safety of the medical treatment in preclinical trials on a small number of participants. This is an essential step in clinical trials.
- v. Determine the feasibility of your study, so you don't waste resources and time.
- vi. Provide preliminary data that you can use to improve your chances for funding and convince stakeholders that you have the necessary skills and expertise to successfully carry out the research.

Pre-testing

Find 5-10 people from your target group

Once you have finished designing your survey questionnaire, find 5-10 people from your target group to pretest it. If you cannot get people from your exact target group then find people who are as close as possible. I once designed a survey that was going to be completed by garment factory workers in another province. There wasn't enough budget available for us to travel to that province to pretest it, so we found some garment factory workers in our own province to test it.

Try to get a range of different people who are representative of your target group. For example, if your target group is young people aged 15-25, try to include some who are younger, some who are older, boys and girls with different socioeconomic backgrounds.

Although 5-10 people might not sound like many, you will usually find that most of them have the same problems with the survey. So even with this small number of people you should be able to identify most of the major issues. Adding more people might identify some additional smaller issues, but it also makes pretesting more time consuming and costly.

Ask them to Complete the Survey while Thinking Out Loud

Once you've found your testers, ask them to complete the survey one at a time (they should not be able to watch each other complete it). The testers should complete the survey the same way that it will be completed in the actual project. So, if it is an online survey they should complete it online, if it's a verbal survey you should have a trained interviewer ask them the questions.

While they are completing the survey ask them to think out loud. Each time they read and answer a question they should tell you exactly what comes into their mind. Take notes on everything they say.

"I don't understand this question"

"The option I want isn't available"

"I can't find the next section"

"This is getting boring. Why is it so long?"

"Why is it asking about that? That makes me uncomfortable"

Observe How they Complete the Survey

You should also observe them completing the survey. Look for places where they hesitate or make mistakes, such as the example below. This is an indication that the survey questions and layout are not clear enough and need to be improved. Keep notes on what you observe.

How important are the following issues for you personally?

Circle the number that best represents your opinion on a scale from 1 (extremely unimportant) to 10 (extremely important)

	Extremely unimportant Extremely important									
Protection of endangered species	1	2	3	4	5	6	7	8	9	10
Improving air quality	1	2	3	4	5	6	7	8	9	10
Creating renewable energy sources	1	2	3	4	5	6	7	8	9	10
Improving water quality	1	2	3	4	5	6	7	8	9	10
Reducing the use of nuclear power	1	2	3	4	5	6	7	8	9	10

Make Improvements Based on the Results

Once all the testers have completed the survey review your notes from each session. At this point it is normally clear what the major problems are so you can go about improving the survey to address those

problems. However, if major changes are needed to the questions or structure, it might be necessary to repeat the pretesting exercise with different people before starting the survey.

Data Analysis and Processing

Data initially obtained must be processed or organized for analysis. For instance, this may involve placing data into rows and columns in a table format for further analysis, such as within a spreadsheet or statistical software.

Data Cleaning

Once processed and organized, the data may be incomplete, contain duplicates, or contain errors. The need for data cleaning will arise from problems in the way that data is entered and stored. Data cleaning is the process of preventing and correcting these errors. Common tasks include record matching, deduplication, and column segmentation.

Such data problems can also be identified through a variety of analytical techniques. For example, with financial information, the totals for particular variables may be compared against separately published numbers believed to be reliable. Unusual amounts above or below pre-determined thresholds may also be reviewed. There are several types of data cleaning that depend on the type of data. Quantitative data methods for outlier detection can be used to get rid of likely incorrectly entered data. Textual data spellcheckers can be used to lessen the number of mistyped words, but it is harder to tell if the words themselves are correct.

Descriptive Vs. Inferential Statistics

When analyzing data, such as the marks achieved by 100 students for a piece of coursework, it is possible to use both descriptive and inferential statistics in your analysis of their marks. Typically, in most research conducted on groups of people, you will use both descriptive and inferential statistics to analyze your results and draw conclusions.

1. Descriptive Statistics

Descriptive statistics is the term given to the analysis of data that helps describe, show or summarize data in a meaningful way such that, for example, patterns might emerge from the data. Descriptive statistics do not, however, allow us to make conclusions beyond the data we have analyzed or reach conclusions regarding any hypotheses we might have made. They are simply a way to describe our data. Descriptive statistics are very important because if we simply presented our raw data, it would be hard to visualize what the data was showing, especially if there was a lot of it. Descriptive statistics therefore enables us to present the data in a more meaningful way, which allows simpler interpretation of the data.

For example, if we had the results of 100 pieces of students' coursework, we may be interested in the overall performance of those students. We would also be interested in the distribution or spread of the marks. Descriptive statistics allow us to do this.

Typically, there are two general types of statistic that are used to describe data:

1. *Measures of central tendency* - these are ways of describing the central position of a frequency distribution for a group of data. In this case, the frequency distribution is simply the distribution and pattern of marks scored by the 100 students from the lowest to the highest. We can describe this central position using a number of statistics, including the mode, median, and mean (Students to read and understand specific measures of central tendency).
2. *Measures of spread* - these are ways of summarizing a group of data by describing how spread out the scores are. For example, the mean score of our 100 students may be 65 out of 100. However, not all students will have scored 65 marks. Rather, their scores will be spread out. Some will be lower and others higher. Measures of spread help us to summarize how spread out these scores are. To describe this spread, a number of statistics are available to us, including the range, quartiles, absolute deviation, variance and standard deviation (Students to read and understand specific measures of spread).

When we use descriptive statistics, it is useful to summarize our group of data using a combination of tabulated description (i.e., tables), graphical description (i.e., graphs and charts) and statistical commentary (i.e., a discussion of the results).

2. Inferential Statistics

We have seen that descriptive statistics provide information about our immediate group of data. For example, we could calculate the mean and standard deviation of the exam marks for the 100 students and this could provide valuable information about this group of 100 students. Any group of data like this, which includes all the data you are interested in, is called a population.

A population can be small or large, as long as it includes all the data you are interested in. For example, if you were only interested in the exam marks of 100 students, the 100 students would represent your population. Descriptive statistics are applied to populations, and the properties of populations, like the mean or standard deviation, are called *parameters* as they represent the whole population (i.e., everybody you are interested in).

Often, however, you do not have access to the whole population you are interested in investigating, but only a limited number of data instead. For example, you might be interested in the exam marks of all students in Tanzania. It is not feasible to measure all exam marks of all students in the whole of Tanzania so you have to measure a smaller sample of students (e.g., 100 students), which are used to represent the larger population of Tanzania students. Properties of samples, such as the mean or standard deviation, are not called parameters, but statistics.

Inferential statistics are techniques that allow us to use these samples to make generalizations about the populations from which the samples were drawn. It is, therefore, important that the sample accurately represents the population. The process of achieving this is called sampling. Inferential statistics arise out of the fact that sampling naturally incurs sampling error and thus a sample is not expected to perfectly represent the population. The methods of inferential statistics are; i) the estimation of parameter(s) and, ii) testing of statistical hypotheses.

Common inferential statistical tools are as follows:

1. Correlation Analysis

Correlation and regression analysis are related in the sense that both deal with relationships among variables. Correlation analysis is conducted to determine how the independent variables are correlated with the dependent variable. Correlation refers to a technique used to measure the relationship between two or more variables. When two things are correlated, it means that they vary together. Correlation can either be positive or negative; Positive correlation means that high scores on one are associated with high scores on the other, and that low scores on one are associated with low scores on the other. Negative correlation, on the other hand, means that high scores on the first thing are associated with low scores on the second.

Example is the correlation between body weight and the time spent on a weight-loss program:

- a) If the program is effective, the higher the amount of time spent on the program, the lower the body weight.
- b) Also, the lower the amount of time spent on the program, the higher the body weight.

Pearson r is a statistic that is commonly used to calculate correlations. The correlation coefficient can take values between -1 through 0 to +1. The sign (+ or -) of the correlation affects its interpretation. When the correlation is positive ($r > 0$), as the value of one variable increases, so does the other.

Example; what does a Pearson $r = 0.80$ mean?

Interpretation:

- i. *The numerical value of the correlation coefficient:* Correlation coefficients can vary numerically between 0.0 and 1.0. The closer the correlation is to 1.0, the stronger the relationship between the two variables. A correlation of 0.0 indicates the absence of a

relationship while a correlation coefficient of 0.80, indicates the presence of a strong relationship.

- ii. *The sign of the correlation coefficient:* A positive correlation coefficient means that as variable X increases, variable Y increases, and conversely, as variable X decreases, variable Y decreases.

In other words, the variables move in the same direction when there is a positive correlation.

A negative correlation means that as variable X increases, variable Y decreases and vice versa; the variables move in opposite directions.

2. Regression Analysis

Regression analysis is a statistical tool for the investigation of relationships between variables. Regression analysis involves identifying the relationship between a dependent variable and one or more independent variables. Usually, the researcher seeks to ascertain the causal effect of one variable upon another; *the effect of a price increase upon demand, or the effect of changes in the money supply upon the inflation rate.*

To explore such issues, the researcher assembles data on the underlying variables of interest and employs regression to estimate the quantitative effect of the causal variables upon the variable that they influence. If the model is deemed satisfactory, the estimated regression equation can be used to predict the value of the dependent variable given values for the independent variables.

For purposes of illustration, suppose that we wish to identify and quantify the factors that determine earnings in the labour market. A moment's reflection suggests a myriad of factors that are associated with variations in earnings across individuals;

- Occupation, age, experience, educational attainment, motivation, and innate ability come to mind, perhaps along with factors such as race and gender that can be of particular concern to lawyers.

For the time being, let us restrict attention to a single factor, call it education.

Regression analysis with a single explanatory variable is termed "*Simple regression*". Let E denote education in years of schooling for each individual, and let I denote the individual's earnings in dollars per year. The assumption is that higher values of E tend to yield higher values of I , then, the hypothesized relationship between education and earnings may be written as; $I = a + bE + e$.

Where;

a = a constant amount (what one earns with zero education);

b = the effect in dollars of an additional year of schooling on income, hypothesized to be positive;

and

e = reflecting other factors that influence earnings.

The variable I is termed the "dependent" variable; E is termed the "independent" variable; a is the "constant term" and b the "coefficient" of the variable E . Plainly, earnings are affected by a variety of factors in addition to years of schooling; factors that were aggregated into the " e " term in the simple regression model above.

Multiple regression is a technique that allows additional factors into the analysis separately so that the effect of each can be estimated. It is valuable for quantifying the impact of various simultaneous influences upon a single dependent variable. For purposes of illustration, consider the introduction into the earnings analysis of a second independent variable called "experience." Holding constant the level of education, we would expect someone who has been working for a longer time to earn more. Let X denote years of experience in the labor force and, as in the case of education, we will assume that it has a linear effect upon earnings that is stable across individuals.

The modified model may be written:

$$I = a + bE + gX + e; \text{ where } g \text{ is expected to be positive.}$$

Note: The most commonly used computer software for data analysis in business and social science research is Statistical Package for Social Sciences (SPSS) and Excel.

The Research Report

Research report is a document that contains basic aspects of the research project. A research report involves relevant information on the research work carried out. It may be in form of hand-written, typed, or computerized. In terms of format, there is no one best format for all reports. Format depends on several relevant variables. It is advisable that the student should seek an approved format by the relevant institution under which the research is being done.

However, one must employ a suitable format to create desirable impression with clarity. Report must be attractive. It should be written systematically and bound carefully. A report must use the format (often called structure) that best fit the needs and wants of its readers. Normally, following format based on the critical chapters in a complete research project is suggested as a basic outline, which has sufficient flexibility to meet most situations and specific-university requirements.

1. **Preliminary Pages** – These are the background material pages that are usually paginated using Arabic numbers (i). They include the cover page, declaration, dedication, acknowledgement, table of contents, list of tables, list of figures, definition of terms, list of acronyms/abbreviations and abstract.
2. **Chapter One: Introduction** – Background to the study, statement of the problem, research objectives, research-hypotheses/questions, significance of the study, scope of the study, limitations of the study.
3. **Chapter Two: Literature Review** - Literature review, usually organized and structured on the key variables under the study. It is usually common practice that the review begins with the dependent variables, key theories/models that inform the study followed with a detailed review of the independent variables. The chapter should also present that conceptual framework, a diagrammatic presentation of the variable relationships as conceptualized by the researcher. The chapter should include proper citations indicating sources of key arguments and contentions presented.
4. **Chapter Three: Research Methodology** – Chapter three contains the methodology that is used to carry out the study. At the critical minimum, the chapter should present a detailed explanation of the research design used, the target population covered, the sampling design which presents the sample size and the technique used to select it, the data collection methods and instruments, data analysis and presentation which should explain the key statistical analysis methods/tools used and how the study findings are presented.
5. **Chapter Four: Results and Discussion** – Chapter four contains a detailed presentation of the study findings and their discussion. The structure of the chapter depends on the research objectives and therefore, the student should be sure to present and discuss the findings systematically based on the research objectives. Of importance is to show a clear linkage between the variables covered in the study. The discussion of the study findings should show clear linkage of the results with relevant theory and previous studies. In this chapter, the student should demonstrate deeper understanding of the research findings with detailed inferencing and interpretation of the results.
6. **Chapter Five: Summary, Conclusions and Recommendations** – Chapter five presents the summary of the study, the conclusions and recommendations of the study. It also contains suggested areas for further research. The structure and presentation of the material in the chapter should be based on the research objectives, and should be derived from chapter four.
7. **References** – The section presents the list of all the materials; textbooks, journal articles were used as sources of literature review, usually presented in ascending order.
8. **Appendices** – The appendices section presents key supporting materials such as the research instrument used, letter of approval to conduct the study and any other relevant material deemed to be supportive of the study.

Pagination of the material; preliminary pages, use Arabic numbers (i, ii, iii) while from chapter one through to the end is done using roman numbers (1, 2).

Referencing Format

The approved referencing format for Jomo Kenyatta University of Agriculture and Technology is; *American Psychological Association (APA)*, with the following order:

Name, year, Title, Town, Publisher

1. **Textbooks**

- ✚ **Single Author:** Pegrum, M. (2009). From blogs to bombs: The future of electronic technologies in education. Crawley, W.A: UWA Publishing.
 - ✚ **Two Authors:** O'Donoghue, T., & Clarke, S. (2010). Leading learning: Process, themes and issues in international contexts. London: Routledge.
 - ✚ **3-5 Authors:** Ranzijn, R., McConnochie, K., & Nolan, W. (2009). Psychology and indigenous Australians: Foundations of cultural competence. South Yarra, Vic: Palgrave MacMillan.
 - ✚ **Book with Different Editions:** Howitt, D., & Cramer, D. (2008). *Introduction to research methods in psychology* (2nd Ed.). Harlow, England: FT Prentice Hall.
- 2. Journal Articles**
- ✚ Mokaya, S. O. (2012). Corporate Entrepreneurship and Organizational Performance: Theoretical Perspectives, Approaches and Outcomes. *International Journal of Arts and Commerce*, Vol. 1, Issue 4, pp 133-143.
- Journal Article Free on the Web**
- ✚ Thomas, K., & Bosch, B. (2005). An exploration of the impact of chronic fatigue syndrome and implications for psychological service provision. *E-Journal of Applied Psychology: Clinical Section*, Vol. 1, Issue 3, pp 23-40. Retrieved from <http://ojs.lib.swin.edu.au/index.php/ejap/article/download/4/13> on 12th March 2015.
- 3. Unpublished Thesis/Research Report**
- ✚ Mokaya, S. K. O. (2013). Effects of Corporate Entrepreneurship on Organizational Performance in the Edible Oil Manufacturing Sector in Kenya (Unpublished PhD Thesis). Nairobi: Kenyatta University.

Notes Prepared by:
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